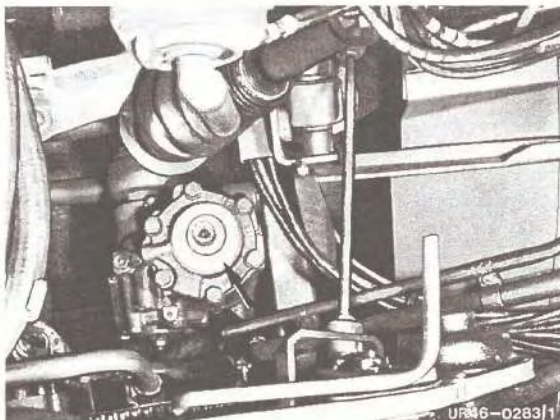
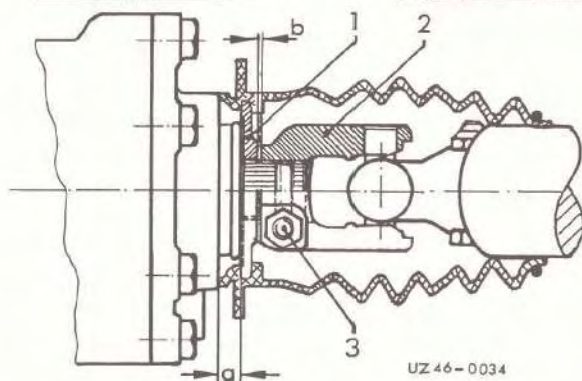


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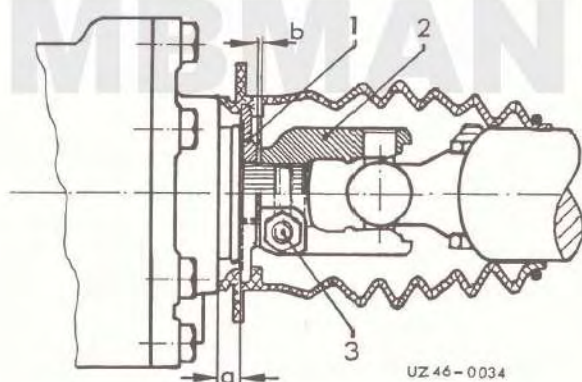
6 Coat supporting ring with grease No. 1 and position on steering box.



7 Slip universal joint into position so that spokes of steering wheel are horizontal when wheels are straight ahead.

8 Check whether clamping bolt (3) of universal joint (2) is located in annular groove around steering shaft.

Note: Dimension „a“ between supporting ring (1) and steering box cover must not exceed 6 to 8 mm. Dimension „b“ between end face of universal joint and supporting ring must not exceed 2 mm.



9 Torque universal joint. See 1.3/1 for tightening torques.

10 Using grease No. 1, pack cavity between bottom edge of supporting ring and cover of steering box. Fasten rubber gaiter.



11 Fill steering system with oil, referring to 1.2/1 for quantity.

12 Raise vehicle at front axle, run engine at idling speed and bleed steering system by turning several times to left and right. Steadily check fluid level.

13 Check wheel lock, correcting if necessary. See 4.1/1.

14 Adjust steering box (steering limiters), referring to 3.1/1.

15 Attach front panel and engine hood.

Checking and adjustment of friction torque

1 Apply special tool No. 7 to steering shaft. Using special tool No. 5, turn steering shaft from stop to stop. Then turn steering shaft back by half the number of turns calculated and check friction torque.
See 1.2/1 for permissible friction torque.

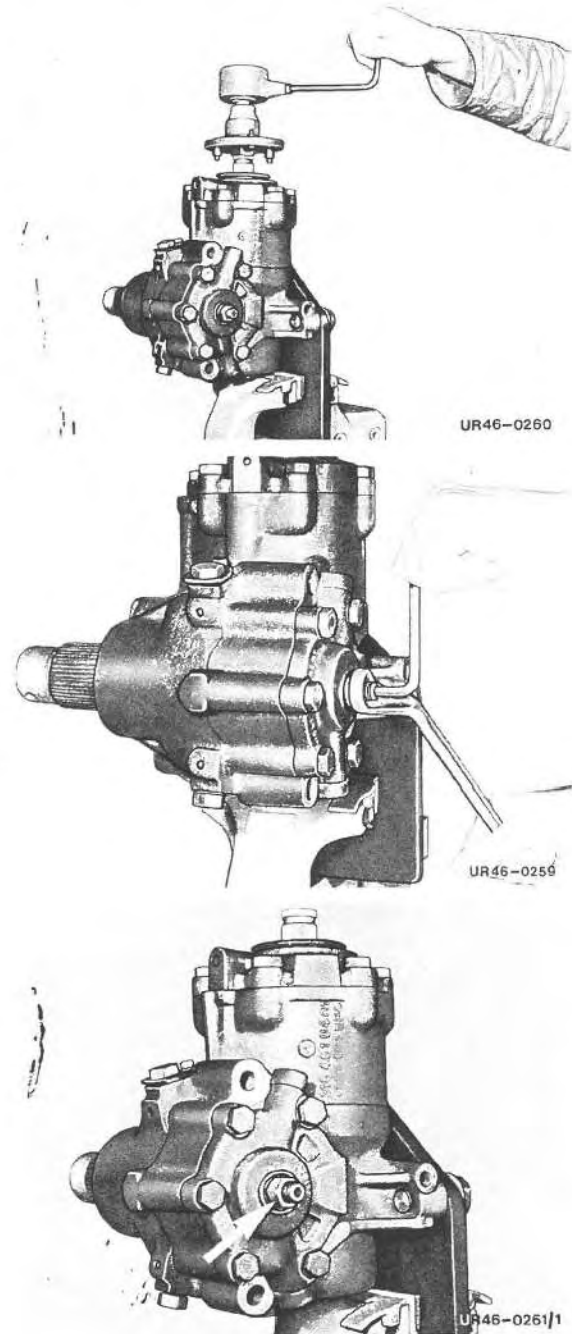
Note:

Marks on steering shaft and end face of bearing cover must agree at center position.

2 If necessary, slacken lock-nut and turn adjusting screw so that friction torque rise of 50 Nm is obtained with steering shaft at center position.

3 Lock adjusting screw and tighten lock-nut.
See 1.3/1 for tightening torques.

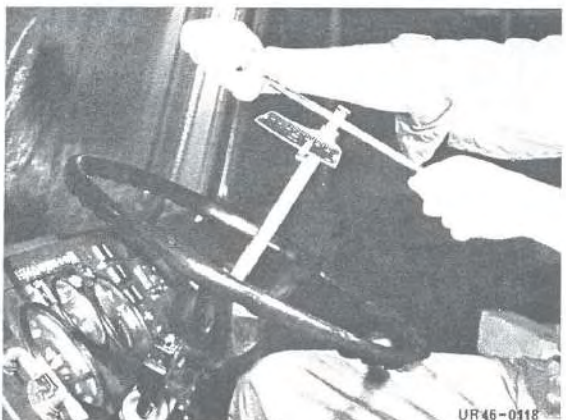
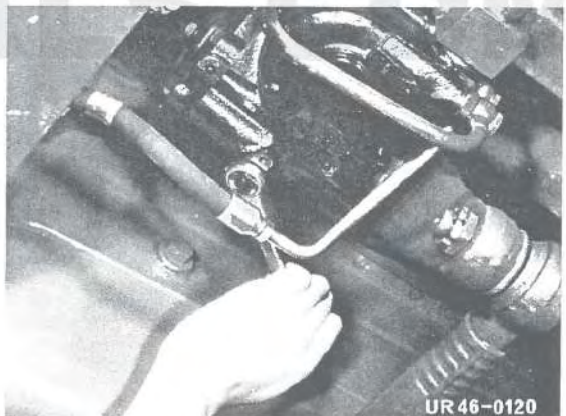
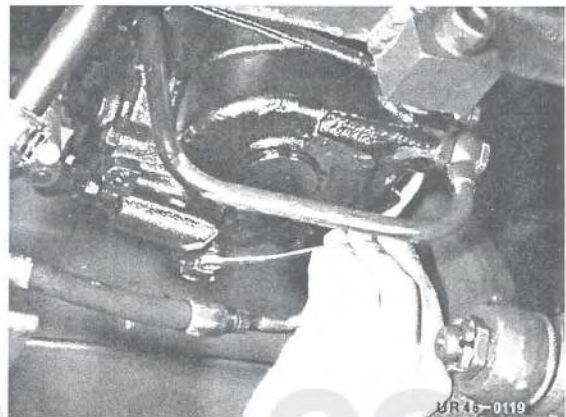
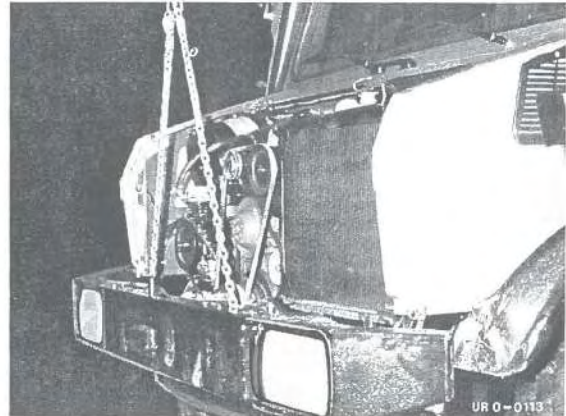
4 Check friction torque again, adjusting with reference to items 1 to 3 if necessary.



Adjustment of steering limiter

Note: For safety reasons make sure that while engine is running, limiter adjusting screws are never unscrewed from steering box by more than 2,5 turns.

- 1 Raise vehicle until both front wheels are clear.
- 2 Remove retaining plate and turn both limiting valves in.
- 3 Limit left-hand lock, while unscrewing left-hand (lower) limiting valve by 2,0 to 2,5 turns.
- 4 Run engine at fast idle and turn steering counter-clockwise to stop.
- 5 Apply torque wrench, turn counter-clockwise to 30 Nm and hold at this position.



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- 6 Slowly turn limiting valve in until torque drops as power assistance begins to take effect.
See 1.3/1 for tightening torques.



- 7 Limit right-hand lock, while unscrewing right-hand (upper) limiting valve by 2,0 to 2,5 turns.
- 8 Carry out operations below with reference to jobs 5 to 7 above, but this time turn steering wheel to right and screw upper limiting valve in.



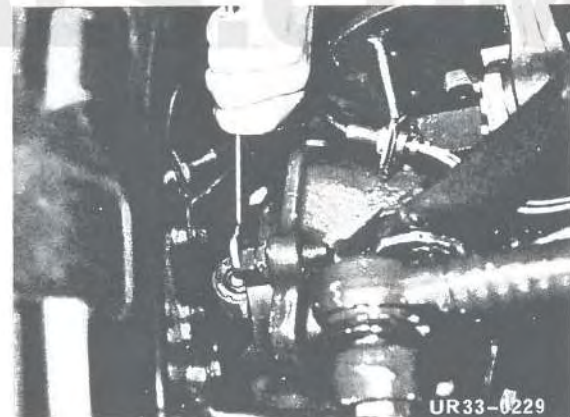
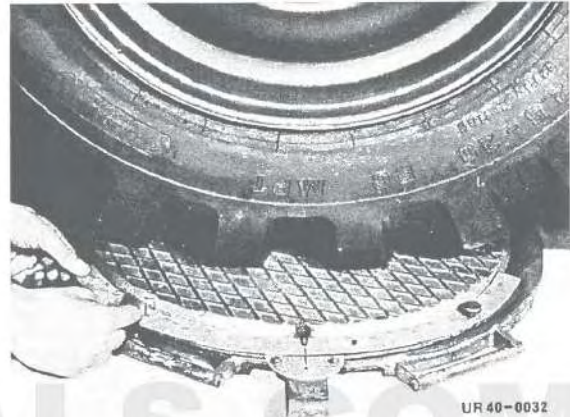
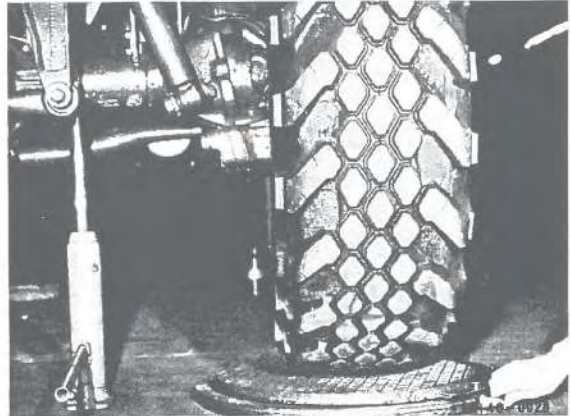
- 9 Bolt retaining plate at correct position and torquing.
See 1.3/1 for tightening torques.

Note: If you fail to attach retaining plate, be sure not to turn limiting valve by more than 150°.

- 10 Check wheel lock, referring to 4.1/1.
- 11 Make sure that gearing wheel fastening nut is seated correctly, torquing to 50 Nm if necessary.

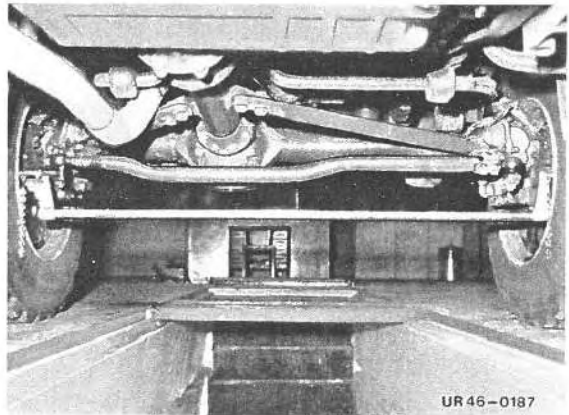
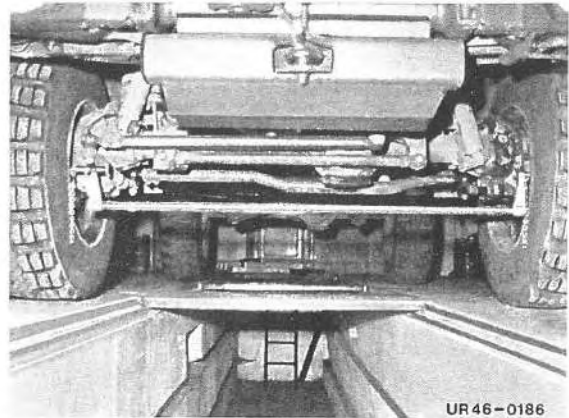
Adjustment

- 1 Park front wheels on testing plates.
- 2 Set front wheels to straight ahead and rotary plate scales to zero.
- 3 Turn steering to left lock, noting angle of left wheel on scale.
- 4 If wheel lock is larger or smaller than 39° to 40° , turn wheel lock bolt in or out accordingly, finally securing again with lock-nut.
- 5 Check right wheel lock in same way, correcting if necessary.
- 6 Jack up vehicle and remove test plates.



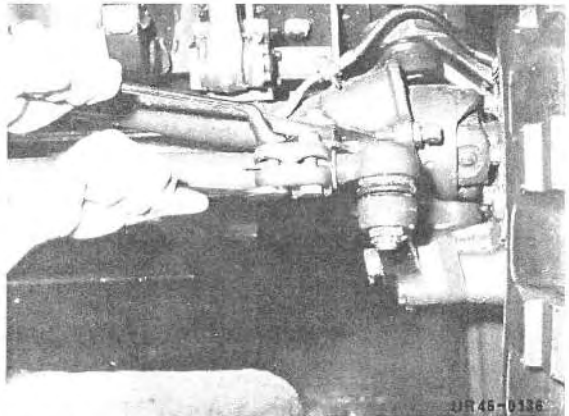
Checking

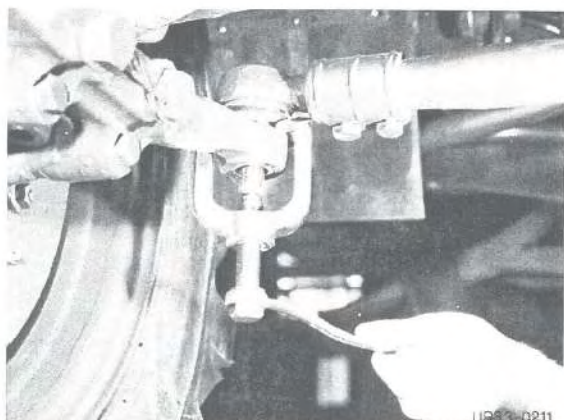
- 1 Park vehicle on flat surface, turning front wheels to straight-ahead position.
- 2 Check tire pressure, correcting if necessary.
- 3 Using wheel alignment tester, check rear wheel track at rim wall on a level with center of wheel.
- 4 Using wheel alignment tester, check rear wheel track at rim wall on a level with center of wheel.
- 5 Examine results, referring to 1.2/1 for correct wheel alignment.



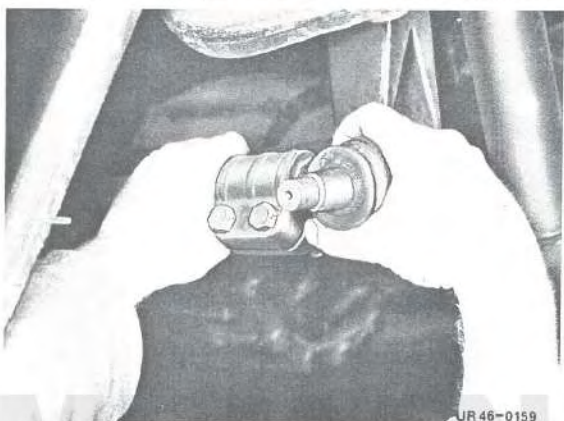
Adjustment

- 1 Release and detach clamp at track rod.
- 2 Release and unscrew castle nut on ball joint.





3 Using special tool No. 2, remove ball joint end.



4 Adjust wheel alignment by screwing ball joint end in or out accordingly.

5 Install in reverse order, referring to 1.3/1 for tightening torque.

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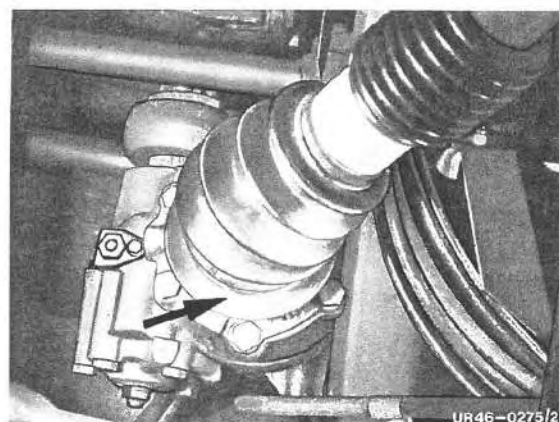
Checking tightness

- 1 Turn front wheels to straight ahead position.
- 2 Raise hood on outside and remove front panel.



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- 3 Pull off rubber seal on steering box and slide up.



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- 4 Press supporting disk (1) against contact face of steering box. Dim. **a** between supporting disk and cover of steering box must not be more than 6 to 8 mm.

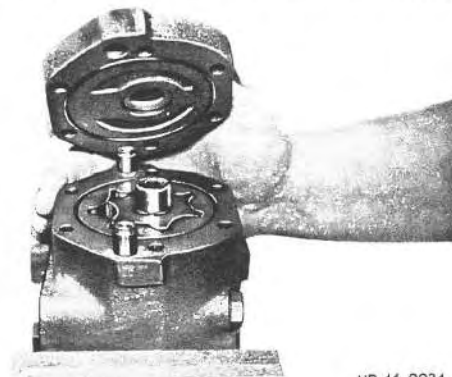
- 5 Check whether universal joint (2) is fully pushed onto splined profile of steering box. Distance from face of universal joint (2) to supporting disk (1), dim. **b**, must not be more than 2 mm.

Important !

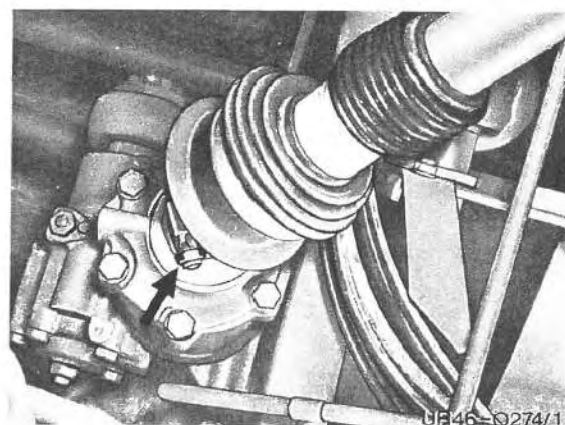
If the universal joint is incorrectly fitted, dim. **b** is 8 mm or more. If this is the case, clamping bolt (3) is not in the annular groove of the steering spindle but runs in front of the splined profile of the steering spindle.

- 6 In this case, unscrew clamping bolt and slide universal joint onto steering spindle to the point where the clamping bolt can be pushed through the annular groove.

- 7 Torque clamping bolt with 64 Nm.

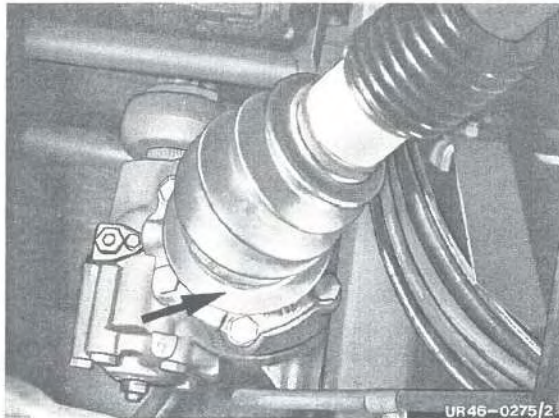


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- 8 Pull over rubber sleeve.
- 9 Fit front panelling and lower bonnet.

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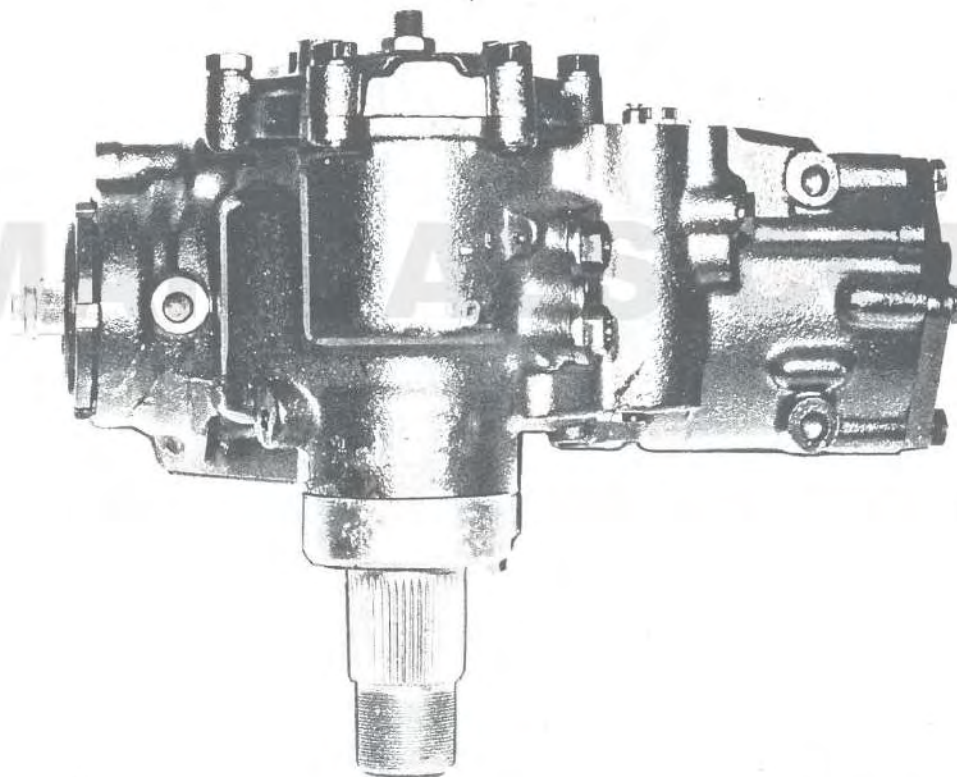
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Installation Survey

Chassis		Steering		Manufacturer	Type	Installation
Model	Sales designation	Model	Sales designation			
435.110	U 1700	765.305	LS 7 F	Daimler-Benz	Worm and nut power steering	Series with SA 35 776
.111	U 1700 L					
.113	U 1700 L/38					
.115	U 1300 L					
.117	U 1300 L/37					
.160	Attachment TM 170					
.170	Attachment Condor					

General view



UR46-0169

Steering gearbox LS 7 F

Technical data**Steering box**

Ratio in steering box		21.78 : 1	
Worm pitch, left-hand	mm	15	
Piston diameter	mm	117	
Swept piston area	cm ²	107,5	
Max. rotation of steering shaft	°	93	
Spline module	mm	10.4	
Pitch radius	mm	52	
Piston stroke	mm	2 x 42.2	
Hydraulic torque	at 100 bar	Nm	5590
	at 120 bar	Nm	6710
Max. torque transmission	Nm	9000	

Steering wheel

Steering wheel diameter	mm	450
Permissible play at steering wheel circumference, with wheel at center position and engine running	mm	40
Steering wheel turns from lock to lock		4.5

Adjustments

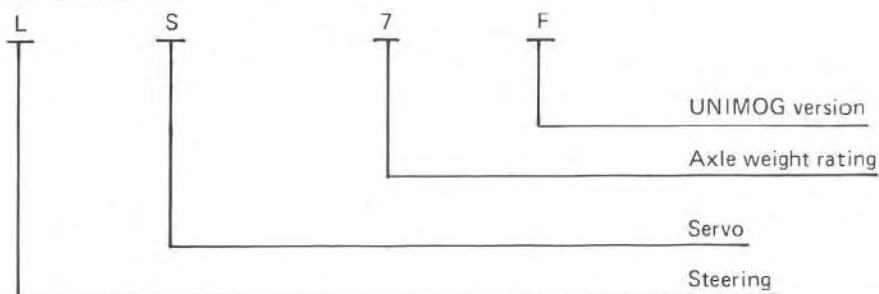
Distance (A) from side member to center of hole in pitman arm	mm	165
Wheel alignment (toe-out)	mm	0 to -4

Capacities

Steering Model designation	Sales designation	Service products ¹⁾	Quantity
765.305	LS 7 F	Engine oil SAE 10 W ²⁾	3,25
		ATF	

¹⁾ Oils listed must not be mixed

²⁾ Use SAE 5 W – 20/30 in cold zones

Steering designation

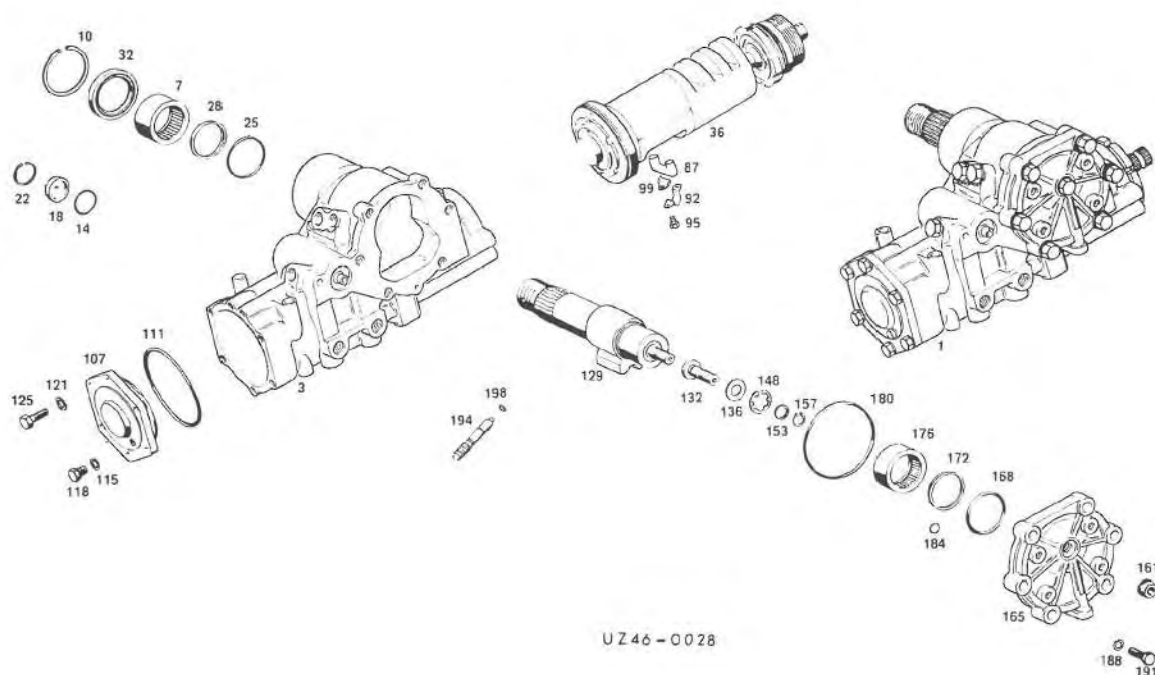
Special tools

Item No.	Designation	Special tool	Tool set
1	Extractor for steering and track rod	405 589 05 33 00	B
2	Extractor for steering wheel (with tool No. 3)	355 589 00 33 00	B
3	Thrust piece	425 589 01 34 00	B

Tightening torques

Designation	Thread	Nm
Steering transmission at frame	M 18 x 1.5 x 65	420
Steering arm and track rod arm at steering knuckle	M 18 x 1.5	400
Pitman arm at steering shaft	M 45 x 1.5	550
Track rod ad tie rod and track rod	M 20 x 1.5	230 ± 20
Steering rod at steering column and pitman arm	M 20 x 1.5	230 ± 20
Clamp at track rod	M 12 x 1.5	90 ± 3
Clamp at steering rod	M 12 x 1.5	90 ± 3
Steering spindle at steering	M 10 x 45	64
Steering wheel at steering spindle	BM 22	50
Mount of steering joint at steering	M 10	65
Mount of steering joint	M 8 x 42	33
Mount of locking strap at steering	M 6 x 12	8 to 10

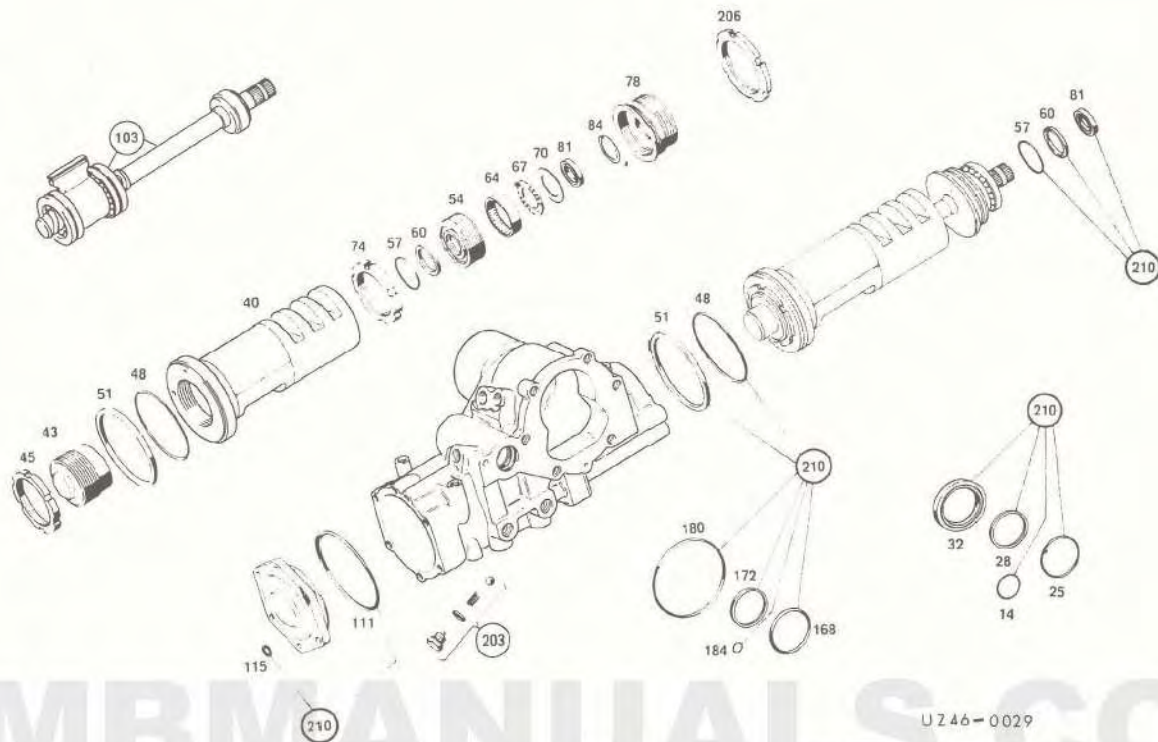
Exploded view



Steering box

1	Steering	118	Screw plug
3	Steering housing	121	Lock washer
7	Needle bearing	125	Bolt
10	Snap ring	129	Steering shaft
14	Sealing ring	132	Bolt
18	Cover	136	Shim
22	Snap ring	148	Retainer
25	Sealing ring	153	Thrust washer
28	Sealing ring	157	Snap ring
32	Sealing ring	161	Nut
36	Piston	165	Cover
87	Ball guide	168	Sealing ring
92	Clip	172	Sealing ring
95	Bolt	176	Needle bearing
99	Retainer	180	Sealing ring
107	Cover	184	Sealing ring
111	Sealing ring	188	Lock washer
115	Sealing ring	191	Bolt

Exploded view



Steering box

14	Sealing ring	74	Nut
25	Sealing ring	78	Screw collar
28	Sealing ring	81	Sealing ring
32	Sealing ring	84	Retainer
40	Piston	103	Repair set
43	Cover	111	Sealing ring
45	Nut	115	Sealing ring
48	Sealing ring	168	Sealing ring
51	Sealing ring	172	Sealing ring
54	Screw collar	180	Sealing ring
57	Sealing ring	184	Sealing ring
60	Sealing ring	203	Repair set
64	Needle bearing	206	Nut
67	Roller bearing	210	Repair set
70	Washer		

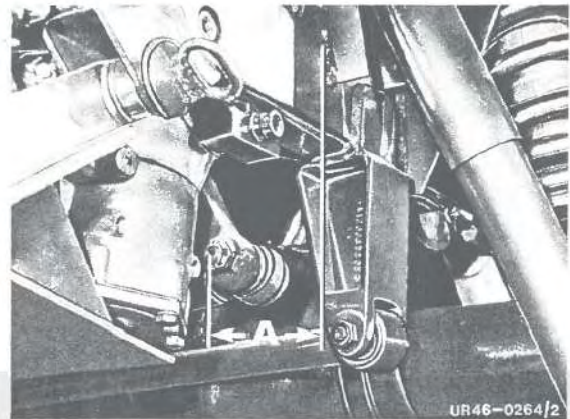
Removal

Note: Prior to removal, move wheels to straight-ahead position and check wheel alignment, referring to 5.1/1.

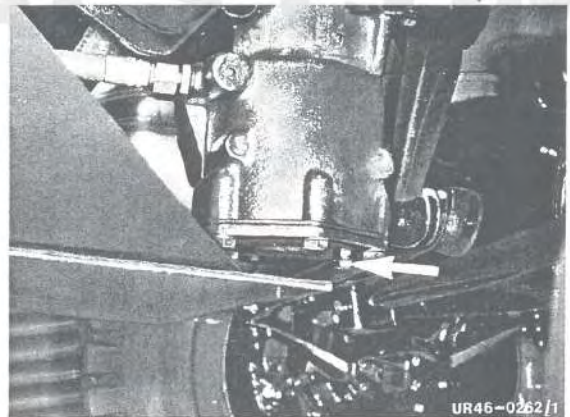
- 1 Remove engine hood and front panel.



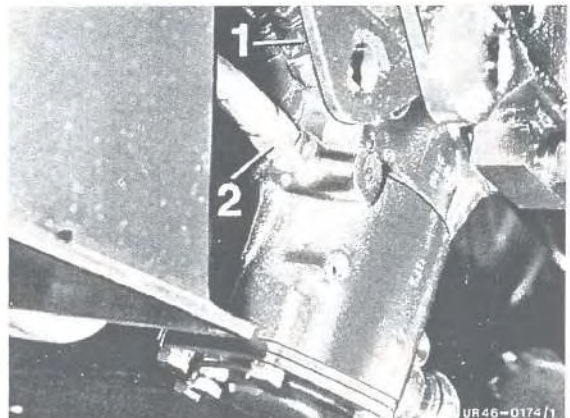
- 2 Unscrew castle nut and remove drag link.



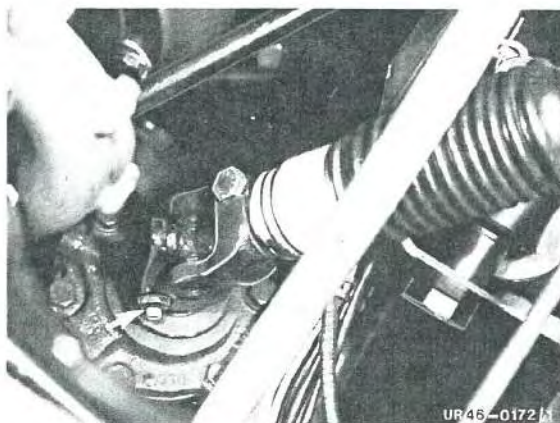
- 3 Drain oil, moving pitman arm so that all oil runs out.



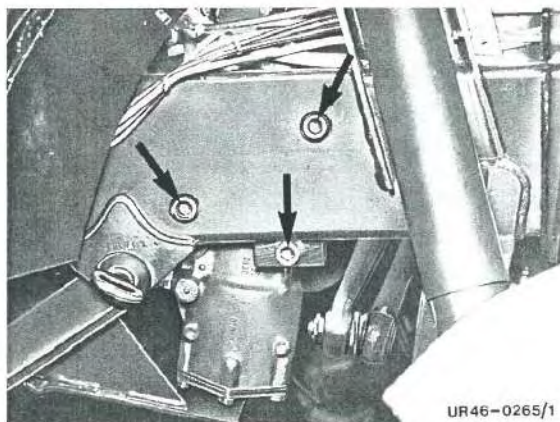
- 4 Unscrew delivery line (1) and return line (2).



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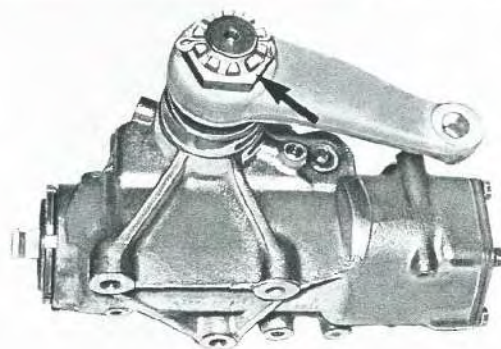


- 5 Detach universal joint at steering box.



- 6 Unscrew and remove steering box.

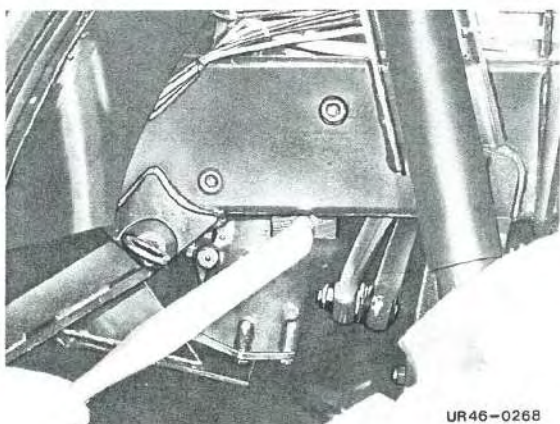
- 7 Release and remove pitman arm.



Installation

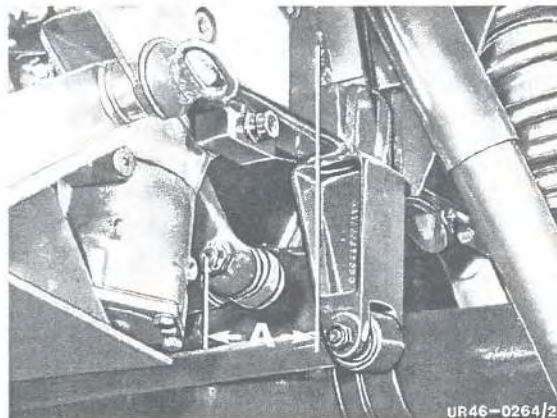
- 1 Attach pitman arm torque and secure.
See 1.3/1 for tightening torques.

Note: Mark on pitman arm must point toward mark on steering shaft.

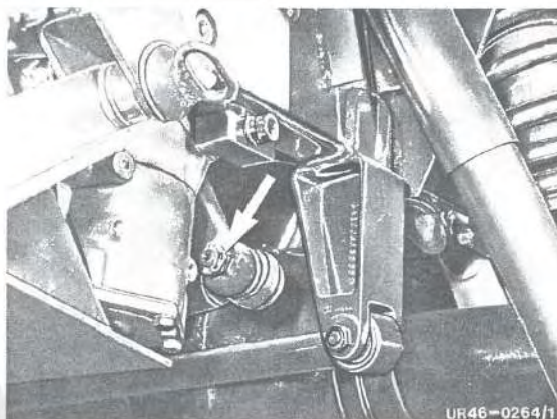


- 2 Attach steering box with conical washer and torque.
See 1.3/1 for tightening torques.

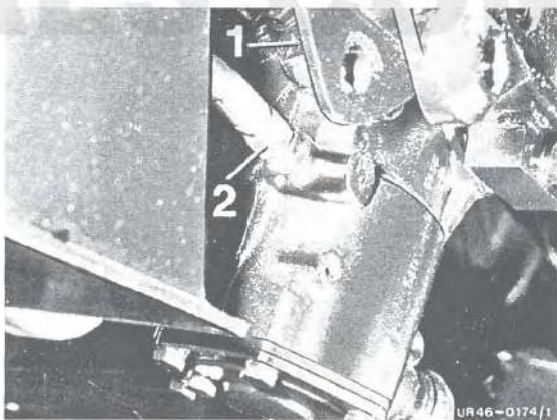
3 Attach drag link. Measure distance between side member and center of hole in pitman arm. If necessary, adjust drag link and track rod so that dimension (A) is obtained. See 1.2/1 for dimension A.



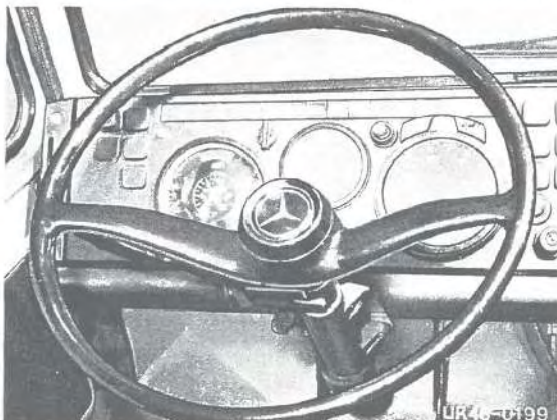
4 Torque drag link and secure. See 1.3/1 for tightening torques.



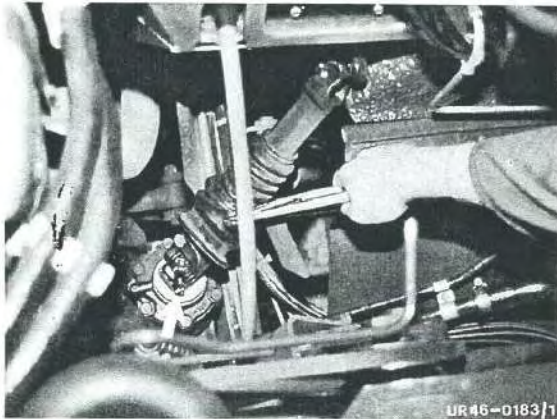
5 Attach delivery line (1) and return line (2).



6 Slip universal joint into position so that spokes of steering wheel are horizontal when wheels are straight ahead.



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- 7 Torque universal joint.
See 1.3/1 for tightening torques.



- 8 Fill system with oil.

- 9 Raise vehicle at front axle, run engine at idling speed and bleed steering system by turning several times to left and right. Steadily check fluid level.



- 10 Check wheel lock, adjusting if necessary with reference to 4.1/1.

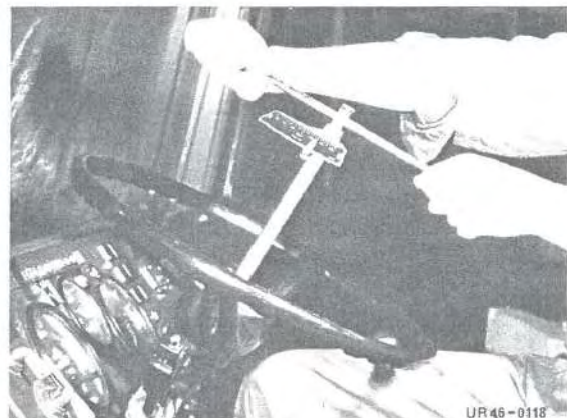
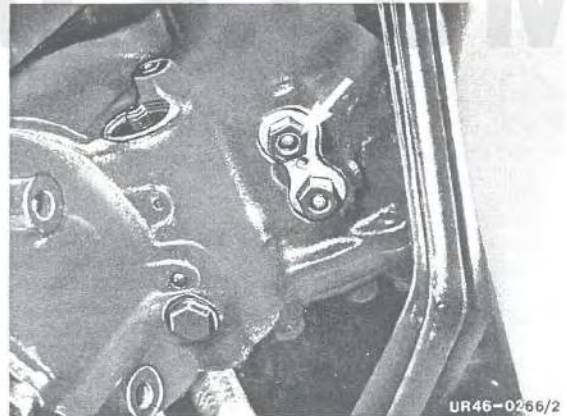
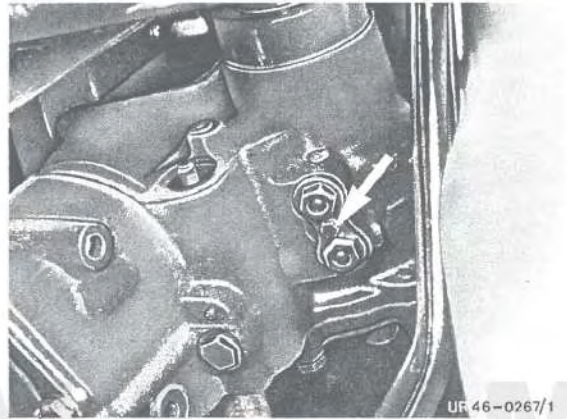
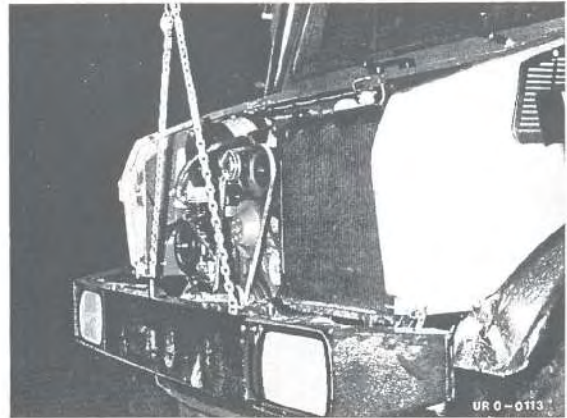
- 11 Adjust steering box (steering limiter), referring to 3.1/1.

- 12 Attach front panel and engine hood.

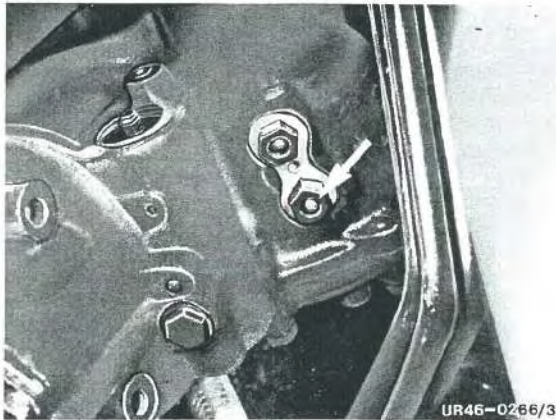
Adjustment

Note: For safety reasons make sure that while engine is running, limiter adjusting screws are never unscrewed from steering box by more than 2,5 turns.

- 1 Raise vehicle until both front wheels are clear.
- 2 Remove retaining plate and turn both limiting valves in.
- 4 Limit left lock, unscrewing front limiting valve by 2,0 to 2,5 turns.
- 5 Run engine at fast idle and turn steering to left lock.
- 6 Apply torque wrench, turn counter-clockwise and hold at this position.
See 1.3/1 for tightening torques.
- 7 Slowly turn limiting valve in until torque drop as power assistance begins to take effect.
See 1.3/1 for tightening torques.

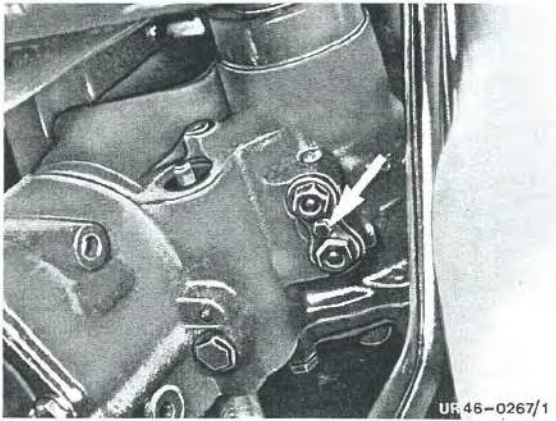


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8 Limit right lock, unscrewing rear limiting valve (2) by 2,0 to 2,5 turns.

9 Carry out operations below with reference to jobs 4 to 6 above, but this time turn steering wheel clockwise and screw rear limiting valve in.



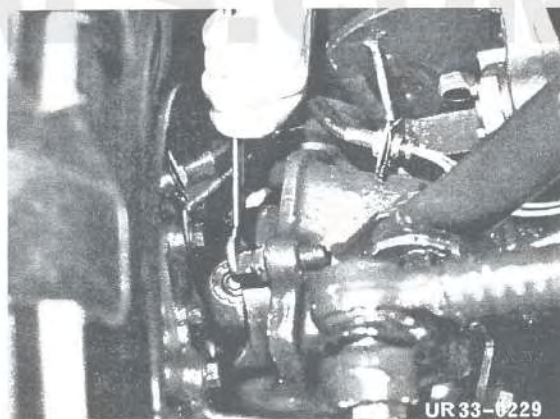
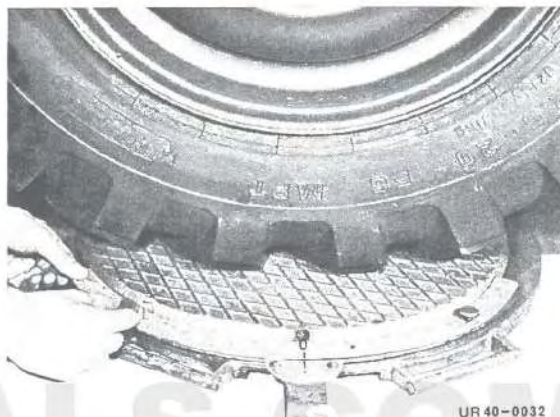
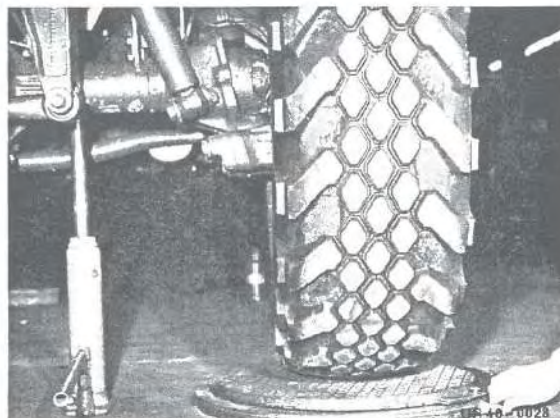
10 Attach retaining plate at correct position and torquing.
See 1.3/1 for tightening torques.

11 Check wheel lock, referring to 4.1/1.

12 Make sure that steering wheel fastening nut is seated correctly, torquing to 50 Nm if necessary.

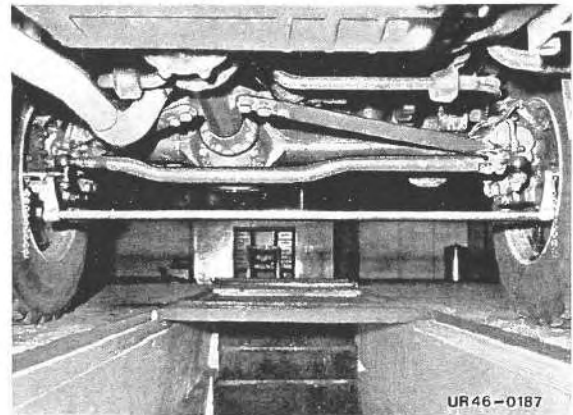
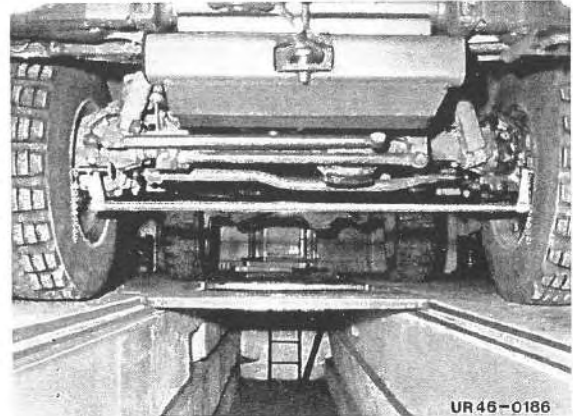
Adjustment

- 1 Park front wheels on testing plates.
- 2 Set front wheels to straight ahead and rotary plate scales to zero.
- 3 Turn steering to left lock, noting angle of left wheel on scale.
- 4 If wheel lock is larger or smaller than 39° to 40° , turn wheel lock bolt in or out accordingly, finally securing again with lock-nut.
- 5 Check right wheel lock in same way, correcting if necessary.
- 6 Jack up vehicle and remove test plates.



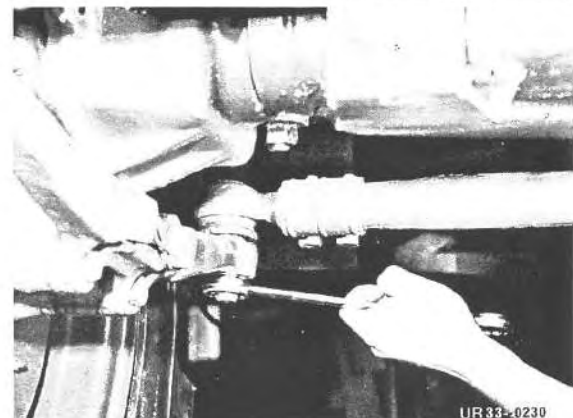
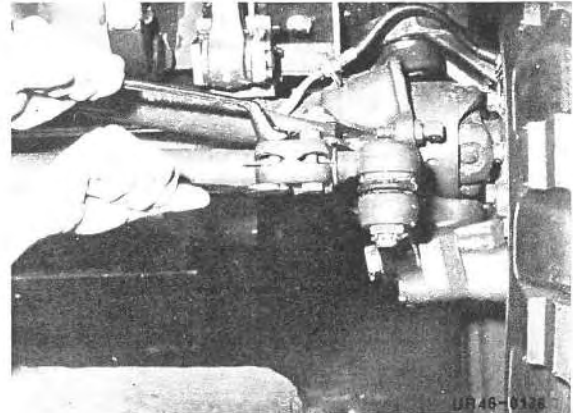
Checking

- 1 Park vehicle on flat surface, turning front wheels to straight-ahead position.
- 2 Check tire pressure, correcting if necessary.
- 3 Using wheel alignment tester, check front wheel track at rim wall on a level with center of wheel.
- 4 Using wheel alignment tester, check rear wheel track at rim wall on a level with center of wheel.
- 5 Examine results, referring to 1.2/1 for correct wheel alignment.



Adjustment

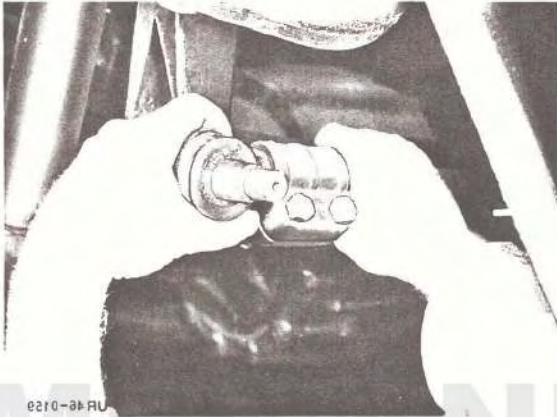
- 1 Release and detach clamp at track rod.
- 2 Release and unscrew castle nut on ball joint.



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3 Using special tool No. 2, remove ball joint end.



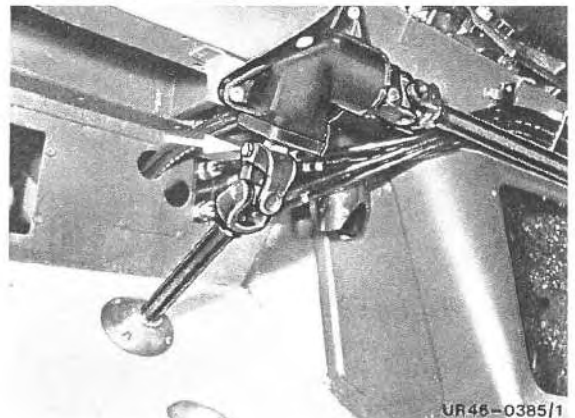
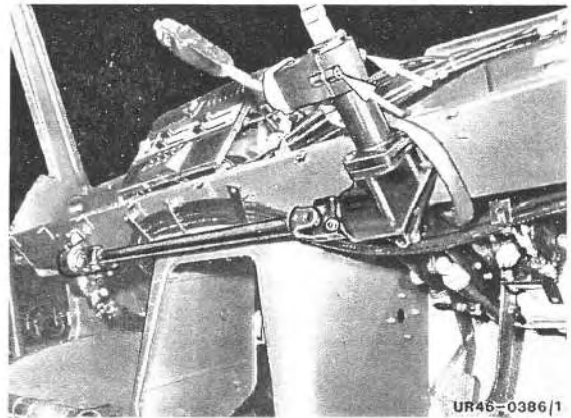
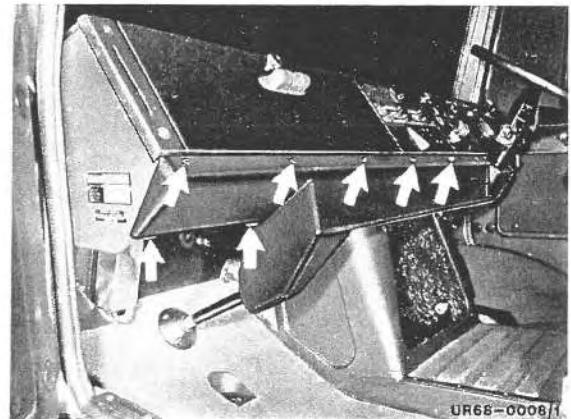
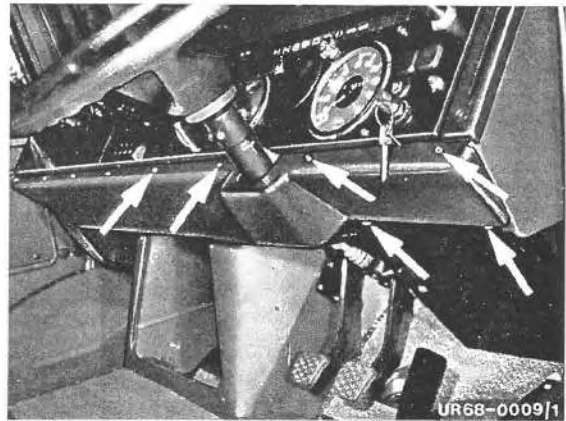
4 Adjust wheel alignment by screwing ball joint end in or out accordingly.

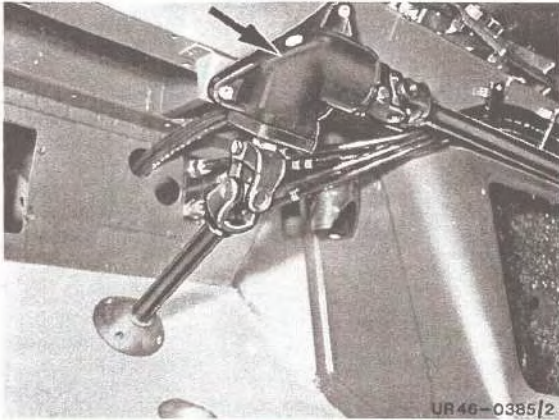
5 Install in reverse order, referring to 1.3/1 for tightening torque.

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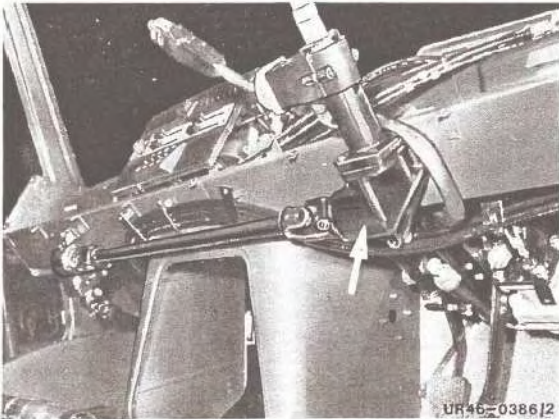
Removal with righthand drive

- 1 Move wheels to straight ahead position.
- 2 Remove lining of intermediate steering shaft at instrument panel.
- 3 Remove lining.
- 4 Remove steering wheel with special tool No. 2.
- 5 Remove combination switch.
- 6 Release universal joint and remove.





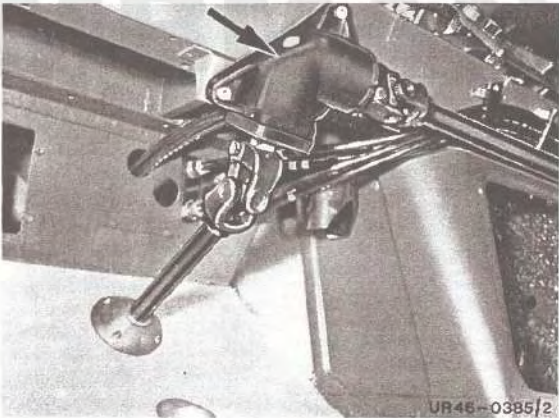
- 7 Remove angle drive left.



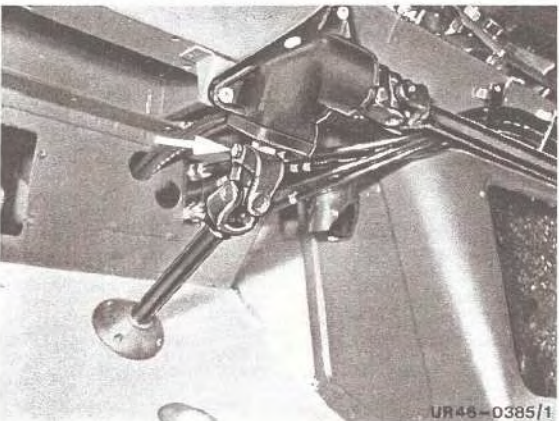
- 8 Remove angle drive right.

Installation with righthand drive

- 1 Fit angle drive right.



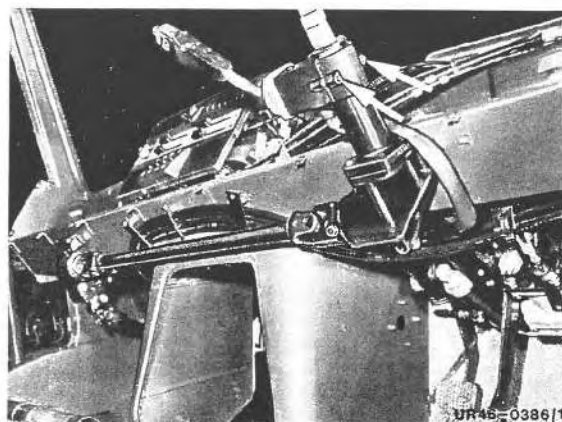
- 2 Fit angle drive left.



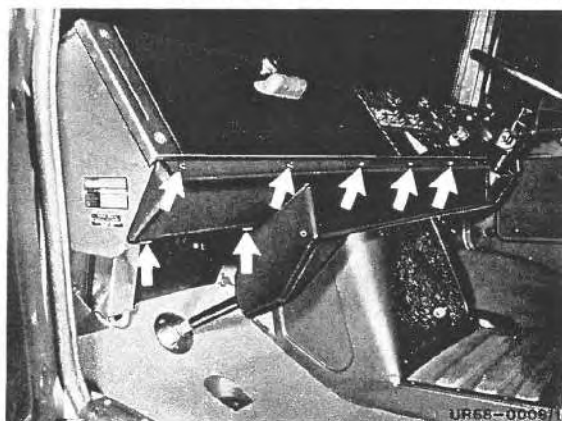
Note: Move wheels to straight ahead position. Before installing universal joint, set return cam of automatic indicator canceller so that it is aligned with the recess on the steering column mount.

- 3 Fit universal joint.

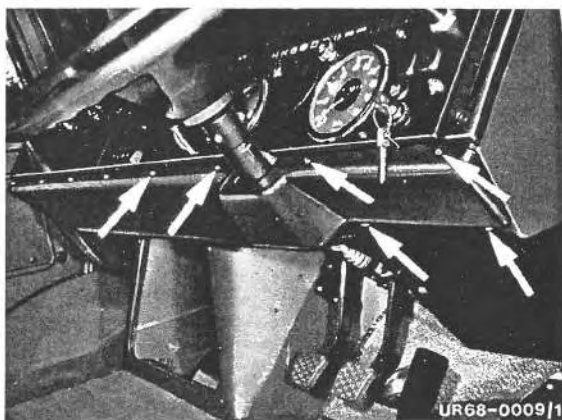
- 4 Fit combination switch.
- 5 Fit steering wheel and tighten.
See 1.3/1 for permissible tightening torques.



- 6 Fit lining.



- 7 Fit lining of intermediate steering shaft.
- 8 Check function of steering.



Contents

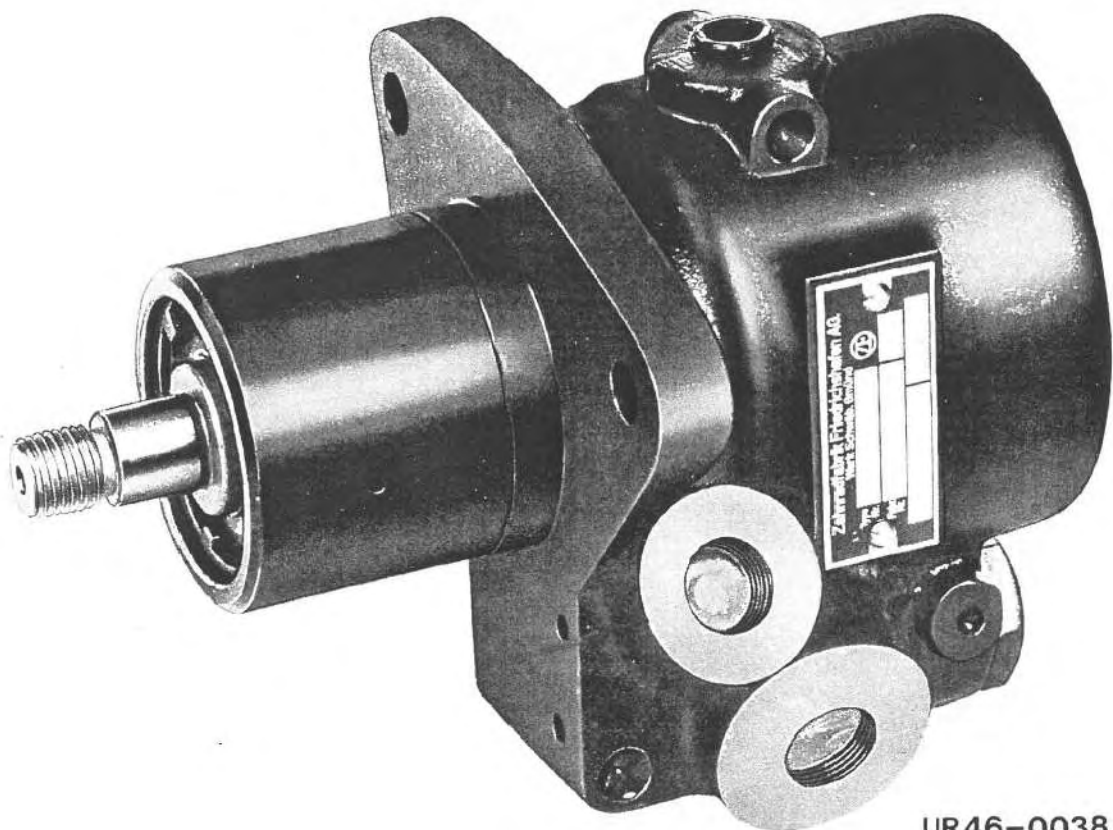
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Exploded view	1.3/1
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Faults and remedies	1.5/1
2 Removal and installation of power steering pump	—
3 Disassembly and assembly of power steering pump	3.1/1
4 Removal, testing and installation of flow limiting valve	4.1/1
5 Checking of pump components	5.1/1

Installation Survey

Chassis		Steering		Hydraulic pump		
Model	Sales designation	Model	Sales designation	Sales designation	Type	Installation
435.110	U 1700	765.305	LS 7 F	ZF 7673 ¹⁾	ZF vane pump with built in flow and pressure limiting valve	Series with SA 35 776
.111	U 1700 L					
.113	U 1700 L/38					
.115	U 1300 L					
.117	U 1300 L/37					
.160	Attachment TM 170					
.170	Attachment Condor					

1) DB part No. 001 466 05 01

General view



UR46-0038

ZF vane pump 7673

Technical data

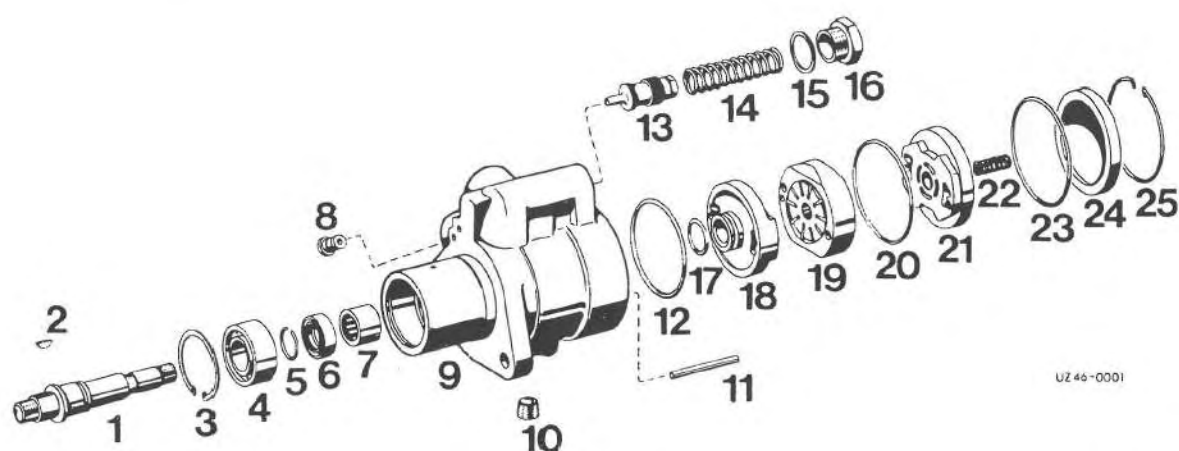
Controlled delivery	Q		L/min	16
	at engine speed	n	1/min	780 to 870
	and at pump speed	n	1/min	900 to 1000
	at pump pressure	p	bar	100
	at oil temperature	t	°C	50 to 60
Max. pressure	p max		bar	100
Speed	n max	at engine speed 2800/min	1/min	3200
Pump rotation				clockwise
Pump drive	Pulley dia.	d	mm	164,5
	V-belt		mm	12,5 x 955 long

Oil grades

Outdoor temperatures		Viscosity	
Zone	Remarks	Summer	Winter
Moderate		SAE 10 W ATF *)	SAE 10 W ATF *)
Cold	Essential to change oil in summer and winter (does not apply to ATF)	SAE 10 W ATF *)	SAE 5 W - 20 H/30 ATF *)
Hot	Temperature in oil tank must not exceed 65°C	SAE 30	SAE 30

*) Applies only to steering

Exploded view



UZ 46-0001

ZF vane pump 7673

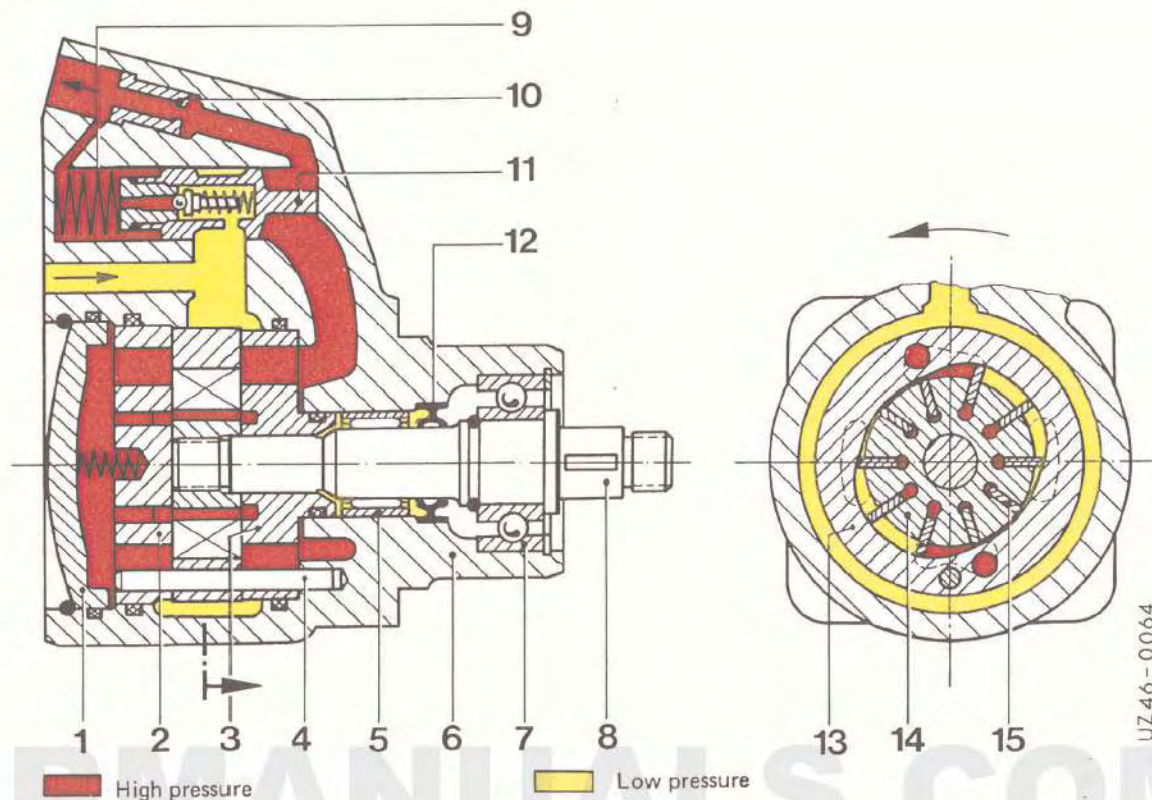
1 Drive shaft	9 Body	17 O-ring
2 Woodruff key	10 Screw plug	18 Face plate, drive end
3 Snap ring	11 Dowel pin	19 Cam ring
4 Deep-groove ball bearing	12 O-ring	20 O-ring
5 Snap ring	13 Valve piston assy.	21 Face plate, cover end
6 Snap ring	14 Compression spring	22 Compression spring
7 Needle bearing	15 Sealing ring	23 O-ring
8 Throttle insert	16 Screw plug	24 Cover
		25 Hooked snap ring

) Flow
limiting
valve

ZF special tools

Cons.No.	Tool	Number	Tool set
1	Fitting drift	7672 798 051	—
2	Fitting drift	7672 798 052	—

Sectional view



ZF vane pump 7673

- | | | |
|-------------------------|----------------------------|----------------------|
| 1 Cover | 6 Housing | 11 Valve piston assy |
| 2 Face plate, cover end | 7 Deep-groove ball bearing | 12 Shaft seal |
| 3 Face plate, drive end | 8 Drive shaft | 13 Cam ring |
| 4 Dowel pin | 9 Compression spring | 14 Rotor |
| 5 Needle bearing | 10 Throttle insert | 15 Blade |

Design and operation

As the drive shaft rotates with the connected rotor, the vanes are thrown radially outwards in slots under the action of centrifugal force and hydraulic oil, being pressed against the guide track of the cam ring. Every two adjacent vanes form a pocket (10 in all) which is confined laterally by pressure plates; every pocket delivers the max. effective pocket volume twice per revolution. The suction and delivery chambers are arranged in such a manner that the hydraulic radial forces acting on the rotor are cancelled mutually.

The face plates at the cover and drive ends have four grooves each, located so that the force of the oil can bear on the inner surface of the pump vanes and thus back up the centrifugal force. Having left the delivery chambers, the oil passes through holes to the flow limiting valve and via a choke to

the delivery line. The reduced pressure downstream of the choke is directed via holes to the spring-loaded end of the flow limiting piston.

As speed and pump delivery rise, the pressure drops still further downstream of the choke, and tends at the spring-loaded end of the valve piston. As soon as the hydraulic force (pressure difference times piston area) exceeds the spring force, the piston moves toward the spring so that the excess oil delivery can flowthrough the opening injection hole and into the pump intake duct. This principle provides a practically constant oil flow for the steering system across the entire speed range. The pressure limiting piston features a spring-loaded ball valve which controls the max. pressure in the pump.

Faults and remedies

Fault	Cause	Remedy
Pump noisy.	Inadequate oil quantity in system (air being sucked in)	While engine is running, raise oil level to upper mark on dipstick and then bleed system.
	Pump sucks air in.	Tighten suction line connections. Renew sealing ring if necessary.
Pressure fails.	Valve piston sticking.	Clean valve piston and check for freedom of movement in valve bore. Fit new valve if necessary. Note tolerance stage.
	Face plate at cover end does not contact cam ring. (Face plate sticking or foreign matter in assembly).	Clean face plate and check for scoring.
	O-rings damaged or forgotten.	Insert new O-rings.
Maximum pressure setting not reached.	V-belt slack or defective.	Tighten or renew V-belt. Thumb test.
	Flow limiting valve fails to close.	Remove and clean flow limiting valve. Throttle ports must not be clogged. Use new valve if necessary. Note tolerance group.
	Pump vanes or cam ring worn.	Fit new set of pump vanes together with new cam ring and rotor. Contact surfaces of face plates at cover and drive ends are also to be checked for wear and damage. Exchange if necessary.
	Rotor shows too much end play.	Renew rotor, cam ring and vanes jointly.
	Foreign matter at face plate.	Clean face plate and check for scoring.
	Flow limiting valve maladjusted.	Exchange flow limiting valve assy. Note tolerance stages.
	Throttle port clogged or pressed in (maximum pressure reached only at very low delivery).	Clean throttle insert.
	O-ring damaged.	Exchange O-ring.

ZF 7673

Fault	Cause	Remedy
Specified delivery not attained.	V-belt slack or damaged.	Tighten or replace V-belt. Thumb test.
	Valve piston sticking or spring is too weak.	Clean piston of flow limiting valve and check for freedom of movement in valve bore. If necessary fit new spring or new valve. Note tolerance group.
	Flow limiting valve fails to close.	Remove piston of flow limiting valve. Release valve screw and clean all parts. Fit new valve if necessary. Note tolerance group.
	Throttle port clogged..	Clean throttle insert.
	Pump vanes or cam ring worn.	Fit new set of pump vanes together with new cam ring and rotor. Contact surfaces of face plates at cover and drive ends are also to be checked for wear and damage. Exchange if necessary.
Pump delivers too much oil.	Rotor shows too much end play.	Renew rotor, cam ring and vanes jointly.
	Valve piston sticking.	Find obstruction and rectify. Foreign matter may be responsible. Replace valve if necessary. Note tolerance group.
	Compression spring too hard.	Replace compression spring.
Max. pressure exceeded.	Bore in flow limiting valve clogged.	Clean parts; fit new valve if necessary. Check max. pressure.
	A washer was forgotten after valve cleaning.	Fit washer. Check max. pressure. Fit new valve if necessary.

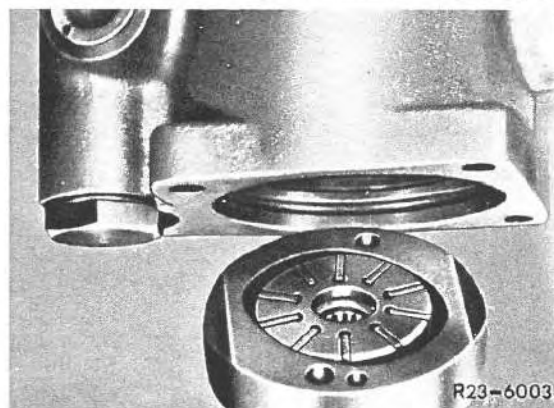
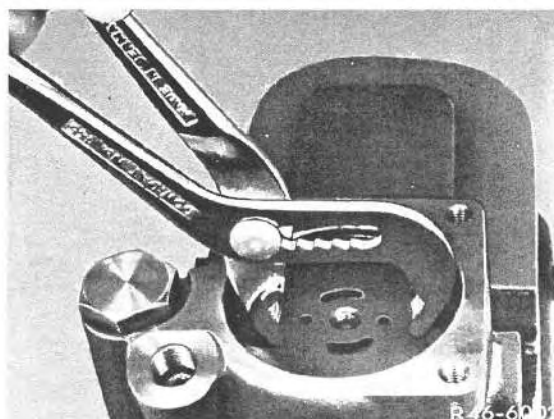
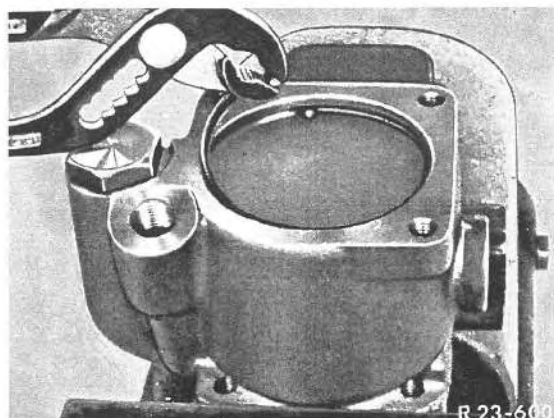
General note:

The following description has been taken over from ZF, together with special tools mentioned here. Special tools marked ZF may be purchased straight from ZF Schwäbisch Gmünd. These operations are only to be executed if replacement of assembly is impossible.

The following illustrations do not show the body as it actually is. Please consult the exploded view as per 1.3/1 for this purpose.

Disassembly**Removal of rotor and cam ring**

- 1 Release nut on drive shaft, draw pulleys off and remove driver.
- 2 Remove hooked snap ring.
- 3 Take cover, compression spring and face plate out of body.
- 4 Tip cam ring and rotor out.

**Removal of drive shaft**

- 5 Prise snap ring out.

